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विद्युत अभियांत्रिकी विभाग



Electronics Lab

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Experiment Number 2

प्रयोग संख्या 2

Verification of OPAMP (IC 741) as an inverter, inverting amplifier, non-inverting amplifier, buffer etc.

इनवर्टर, इनवर्टिंग एम्पलीफायर, नॉन-इनवर्टिंग एम्पलीफायर, बफर आदि के रूप में OPAMP (IC 741) का सत्यापन।

STUDY OF OP-AMP IN VARIOUS CONFIGURATIONS

Aim: To study the Op-Amp in various configurations (as an inverting amplifier, non-inverting amplifier, summing amplifier, buffer etc.

Instruments required:

1. Linear IC Trainer kit
2. Power supply
3. Multimeter

List of experiments:

1. OP-AMP AS INVERTING AMPLIFIER
2. OP-AMP AS NON-INVERTING AMPLIFIER
3. OP-AMP AS UNITY GAIN AMPLIFIER (BUFFER)

Theory:

APPLICATIONS OF OPERATIONAL AMPLIFIER:

The operational amplifier is a versatile device that can be used to amplify both ac and dc input signals and was originally designed for computing such mathematical functions as addition, subtraction, multiplication, and integration. With the addition of suitable external feedback components, the modern-day op-amp can be used for a variety of applications, such as ac and dc signal amplification, active filters, oscillators, comparators, regulators, and others.

An ideal op-amp exhibit the following electrical characteristics:

1. Infinite voltage gain A .
2. Infinite input resistance R , so that almost any signal source can drive it, and there is no loading of the preceding stage.
3. Zero output resistance R_O so that output can drive an infinite number of other devices.
4. Zero output voltage when output voltage is zero.
5. Infinite bandwidth so that any frequency signal can be amplified without attenuation.
6. Infinite common-mode rejection ratio so that the output common-mode noise voltage is zero.
7. Infinite slew rate so that output voltage changes occur simultaneously with input voltage changes.

APPLICATIONS OF IC 555:

With the monolithic integrated circuit 555 we can get accurate timing ranges of microseconds to hours, dependent of supply voltage variations. This versatile device has a large number of interesting practical applications, specially for electronic hobbyists.

Basically, the 555 timer is a highly stable integrated circuit capable of functioning as accurate time-delay generator and as a free running multivibrator, when used as an oscillator the frequency and duty cycle are accurately controlled by only two external resistors and a capacitor. The circuit may be triggered and reset on falling wave rms.

Its prominent features are:-

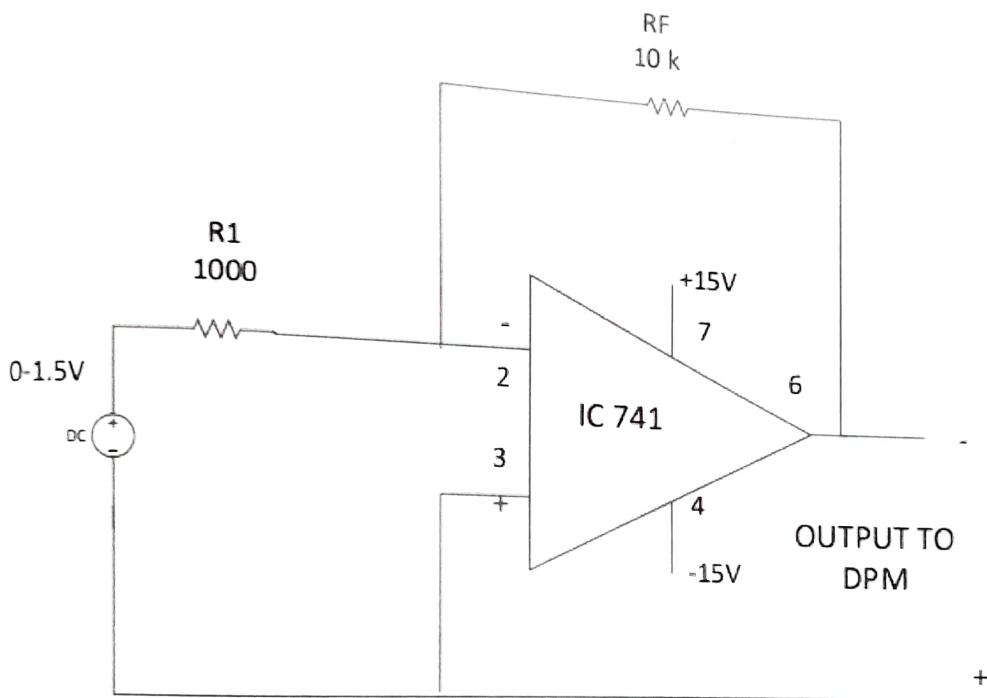
- 1) Timing from micro seconds through hours.
- 2) Monostable and astable operation.
- 3) Adjustable duty cycle.
- 4) Ability to operate from a wide range of supply V_O
- 5) Output compatible with CMOS, DTL & TTL
- 6) High current output can sink and source 200mA
- 7) Trigger and reset inputs are logic compatible.
- 8) Output can be operated normal on and normal off
- 9) High temperature stability.

EXPERIMENT 2.1

Aim: To study Op-Amp as inverting amplifier

Procedure:

1. Connect the circuit as shown in the figure.



2. Use R1 (1 K ohm) in the input circuit and Rf (10 k ohm) in the feedback circuit.
3. Set the input voltage (Vin) at 0.5 V
4. Note down the output using de voltmeter or multimeter.
5. Repeat steps 2-4 using different input voltages (0.75 and 1 volts)

Formula for calculation of output voltage:

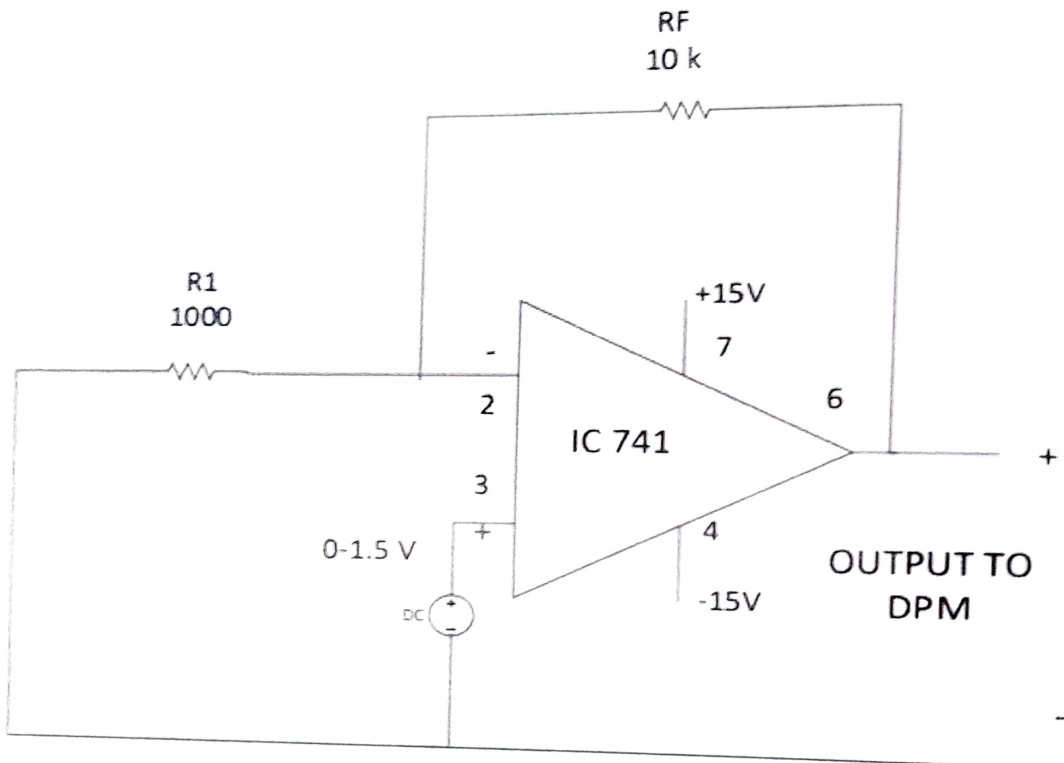
$$V_{out} = -V_{in} (R_f/R_i)$$

EXPERIMENT 2.2

Aim: To study Op-Amp as Non - Inverting

Procedure:

1. Connect the circuit as shown in the figure.



2. Use R1 (1 K ohm) in the input circuit and Rf (10 k ohm) in the feedback circuit.
3. Set the input voltage (Vin) at 0.5 V
4. Note down the output using dc voltmeter or multimeter.
5. Repeat steps 2-4 using different input voltages (0.75 and 1 volts)

Formula for calculation of output voltage:

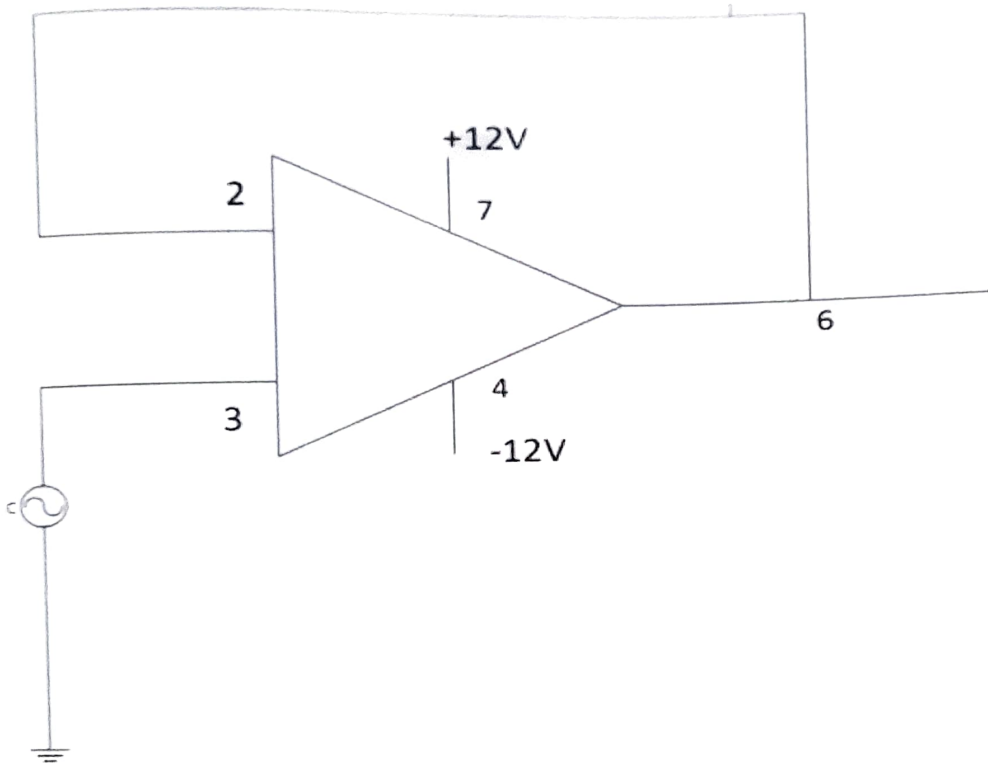
$$V_{out} = V_{in} (1 + R_f / R_i)$$

EXPERIMENT 2.3

Aim: To study Op-Amp as unity gain amplifier (Buffer)

Procedure:

1. Connect the circuit as shown in figure



2. Connect $R1 = 1\text{ k}\Omega$ on pin no 2 and connect the power supply 0-1.5 V DC to Non-inverting input pin no 3 and negative terminal to power supply to ground.
3. For feedback connect pin no 6 directly to pin no 2.
4. Vary the input supply voltage (0- 1.5V Dc) in small steps and note down the output through analog meter:

$$V_{out} = V_{in} \text{ (As gain} = 1\text{)}$$

Result & Conclusion:

Questions:

1. How does the inverting amplifier configuration affect the phase of the output signal?
2. What is the effect of the resistor values on the gain and bandwidth of a non-inverting amplifier?
3. How does the summing amplifier produce an output that is the sum of the inputs?
4. What is the primary function of the buffer configuration in a circuit?
5. What are the typical applications of a differential amplifier in electronic circuits?
6. How does the integrator circuit work to produce an output that is proportional to the integral of the input?